

Retail Customer Behavior & Revenue Analytics

End-to-End Demographic Segmentation & Sales Performance

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Domain: Retail & Consumer Goods

Core Tech Stack: MySQL, Microsoft Power BI, Power Query

Introduction

- This project analyzes customer shopping data from multiple malls between 2021 and 2023
- The goal is to understand customer behavior, revenue drivers, and business patterns

Project Objective

- Identify which customer segments generate more revenue and volume
- Understand category performance and payment preferences
- Provide practical recommendations based on data insights

Dataset Overview

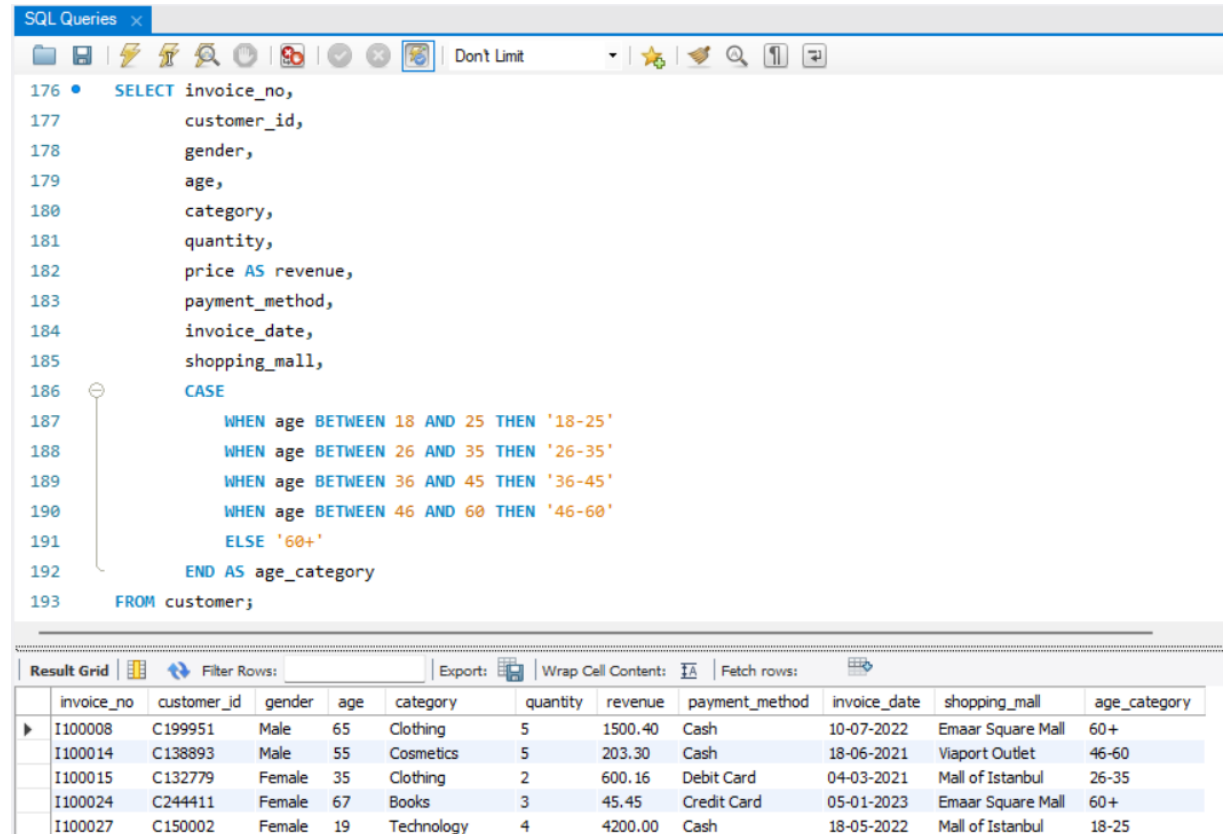
- Dataset includes invoice details, age, gender, category, quantity, price, payment method, and date
- Covers 99K transactions and nearly 299K products sold
- Total revenue recorded is approximately 68.55M

Data Challenges Identified

- Price column represented total transaction revenue instead of unit price
- Raw age values made demographic pattern analysis difficult
- Data restructuring was required before meaningful analysis

Data Transformation in MySQL

- Applied SELECT-only transformation without schema modification
- Renamed price → revenue using alias
- Created age_category using CASE-based segmentation
- Exported transformed analytical dataset



The screenshot shows a MySQL SQL editor window with a query that selects various fields from a 'customer' table and creates a new 'age_category' field using a CASE statement. The query is as follows:

```
176 • SELECT invoice_no,  
177         customer_id,  
178         gender,  
179         age,  
180         category,  
181         quantity,  
182         price AS revenue,  
183         payment_method,  
184         invoice_date,  
185         shopping_mall,  
186         CASE  
187             WHEN age BETWEEN 18 AND 25 THEN '18-25'  
188             WHEN age BETWEEN 26 AND 35 THEN '26-35'  
189             WHEN age BETWEEN 36 AND 45 THEN '36-45'  
190             WHEN age BETWEEN 46 AND 60 THEN '46-60'  
191             ELSE '60+'  
192         END AS age_category  
193 FROM customer;
```

Below the query, the 'Result Grid' shows the output of the query. The results are as follows:

	invoice_no	customer_id	gender	age	category	quantity	revenue	payment_method	invoice_date	shopping_mall	age_category
▶	I100008	C199951	Male	65	Clothing	5	1500.40	Cash	10-07-2022	Emaar Square Mall	60+
	I100014	C138893	Male	55	Cosmetics	5	203.30	Cash	18-06-2021	Viaport Outlet	46-60
	I100015	C132779	Female	35	Clothing	2	600.16	Debit Card	04-03-2021	Mall of Istanbul	26-35
	I100024	C244411	Female	67	Books	3	45.45	Credit Card	05-01-2023	Emaar Square Mall	60+
	I100027	C150002	Female	19	Technology	4	4200.00	Cash	18-05-2022	Mall of Istanbul	18-25

Data Transformation in Power Query

- Renamed price to revenue for reporting consistency
- Created structured age_group bins (18-25 to 60+)
- Extracted Year & Month for time analysis
- Standardized categorical fields

Table: RenameColumns(#"Changed Type",{"price", "revenue"})

category	quantity	revenue	payment_method	invoice_date	shopping_mall
Clothing	5	1500.4	Cash	10-07-2022	Emaar Square Mall
Cosmetics	5	203.3	Cash	18-06-2021	Viaport Outlet
Clothing	2	600.16	Debit Card	04-03-2021	Mall of Istanbul
Books	3	45.45	Credit Card	05-01-2023	Emaar Square Mall
Technology	4	4200	Cash	18-05-2022	Mall of Istanbul
Clothing	1	300.08	Credit Card	19-06-2021	Metrocity
Food & Beverage	3	15.69	Credit Card	23-06-2022	Emaar Square Mall
Food & Beverage	4	20.92	Credit Card	19-10-2021	Zorlu Center
Clothing	1	300.08	Debit Card	06-06-2021	Kanyon
Cosmetics	3	121.98	Cash	24-07-2022	Mall of Istanbul
Shoes	1	600.17	Debit Card	26-12-2021	Mall of Istanbul
Cosmetics	3	121.98	Credit Card	02-01-2023	Metrocity
Clothing	2	600.16	Credit Card	13-10-2021	Forum Istanbul
Souvenir	3	35.19	Debit Card	14-06-2021	Metrocity
Toys	2	71.68	Credit Card	13-08-2021	Metropol AVM
Clothing	3	900.24	Cash	16-10-2021	Mall of Istanbul
Technology	5	5250	Debit Card	10-08-2021	Metropol AVM
Food & Beverage	1	5.23	Credit Card	03-02-2021	Emaar Square Mall
Books	5	75.75	Credit Card	10-07-2021	Mall of Istanbul
Toys	4	143.36	Credit Card	26-04-2022	Metropol AVM

Query Settings

PROPERTIES

Name: customer

APPLIED STEPS

Source, Promoted Headers, Changed Type, **Renamed Columns**

Table: AddColumn(#"Renamed Columns", "age_group", each if [age] >= 18 and [age] <= 25 then "18-25")

quantity	age_group
2	18-25
2	18-25

Custom Column

Add a column that is computed from the other columns.

New column name: age_group

Custom column formula: `= if [age] >= 18 and [age] <= 25 then "18-25"
else if [age] <= 35 then "26-35"
else if [age] <= 45 then "36-45"
else if [age] <= 60 then "46-60"
else "60+"`

Available columns: invoice_no, customer_id, gender, age, category, quantity, revenue

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

Query Settings

PROPERTIES

Name: customer

APPLIED STEPS

Source, Promoted Headers, Changed Type, Renamed Columns, **Custom1**

Overall Performance Summary

- Total Revenue: 68.55M
- Total Transactions: 99K
- Total Quantity Sold: 299K
- Average Order Value: 689.26

68.55M
Total Revenue

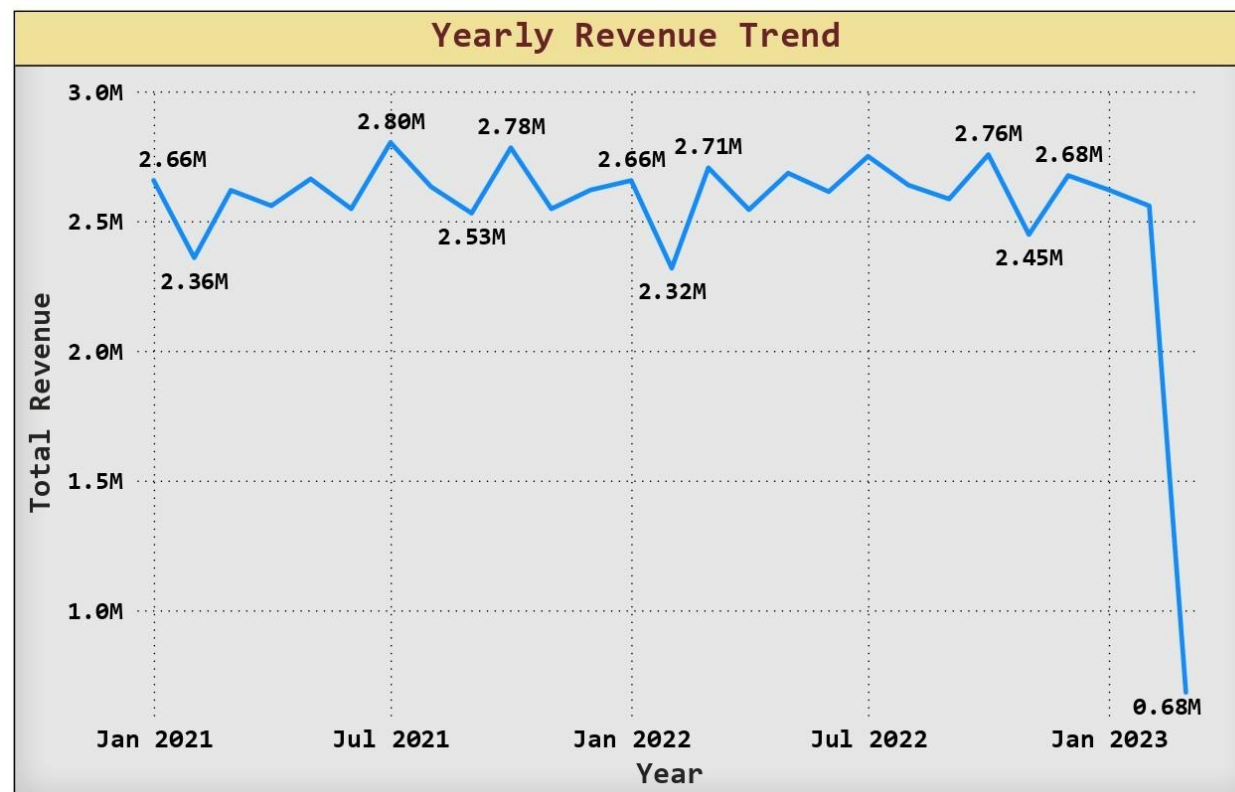
99K
Total Transactions

299K
Total Quantity

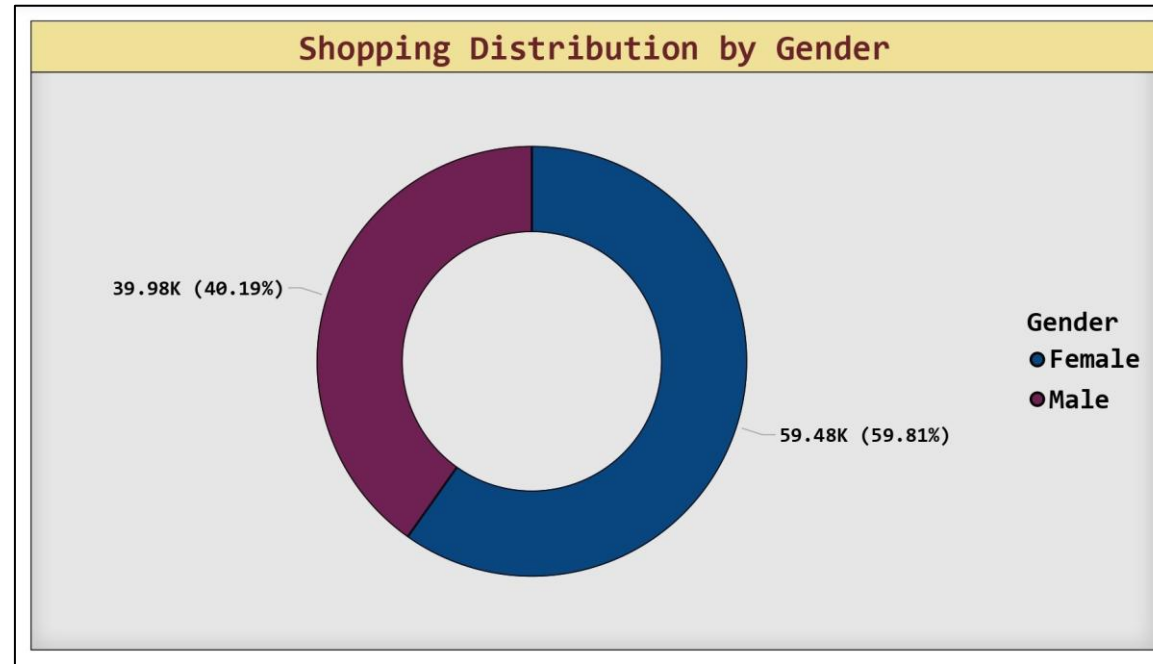
689.26
Average Order Value

Revenue Trend Over Time

- Revenue remained stable across years with consistent monthly performance
- No extreme seasonal fluctuations observed
- Indicates steady customer demand



Shopping Distribution by Gender



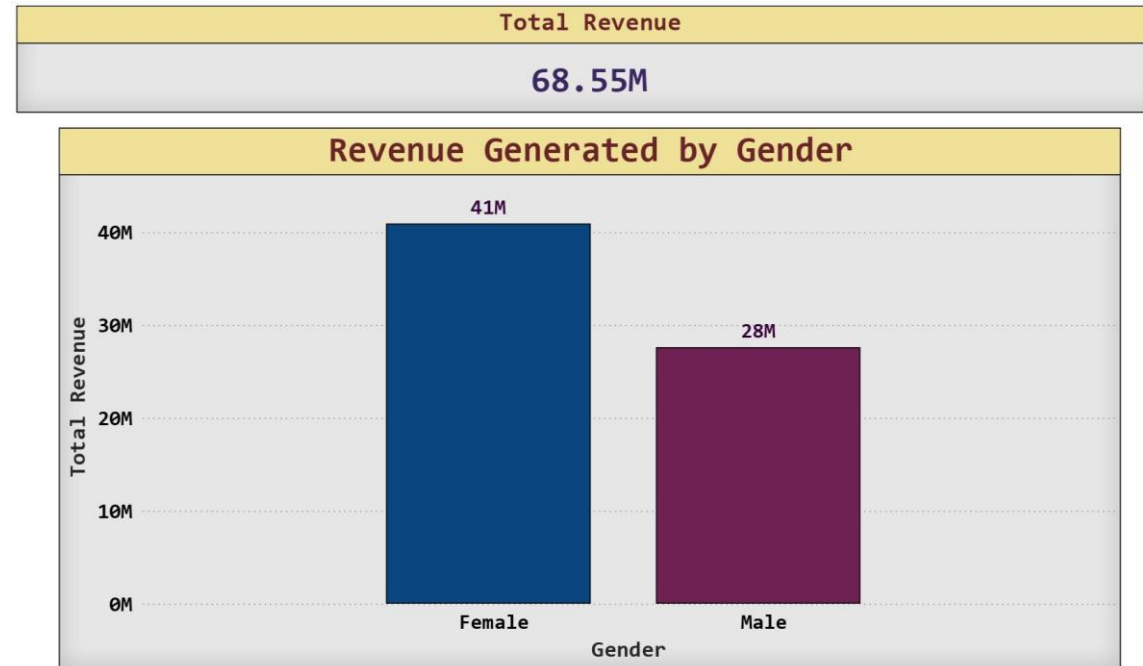
- Female customers contribute around 60% of total transactions
- Male customers contribute around 40%
- Female segment forms the larger customer base

Product Volume by Gender



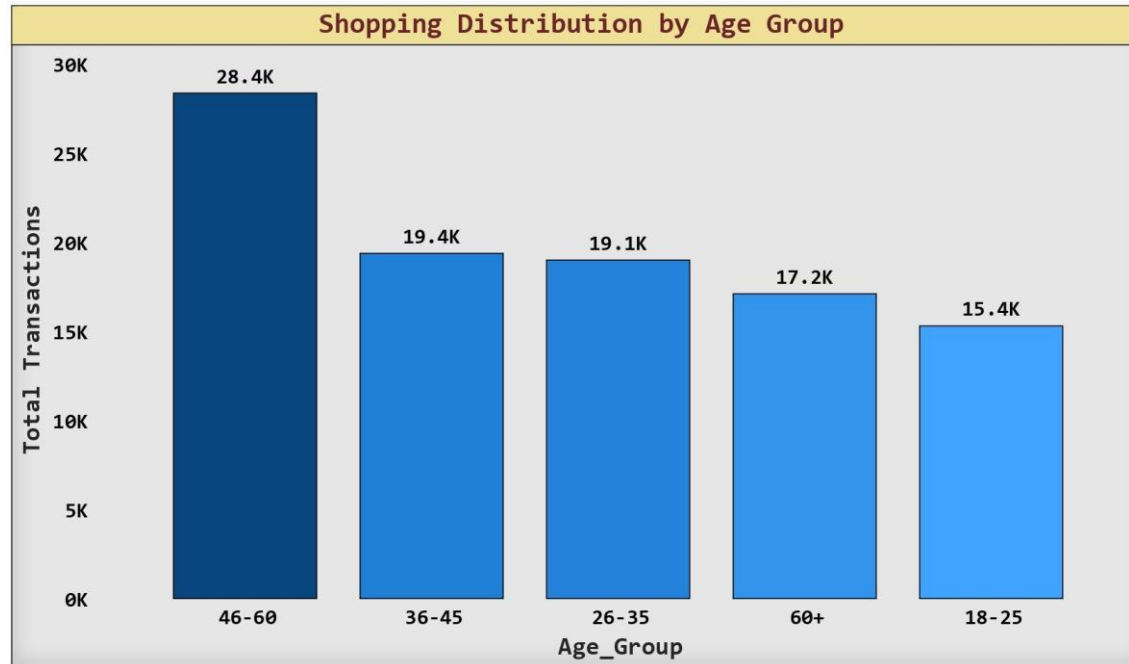
- Females purchased around 179K products
- Males purchased around 120K products
- Female customers show higher purchase volume

Revenue by Gender



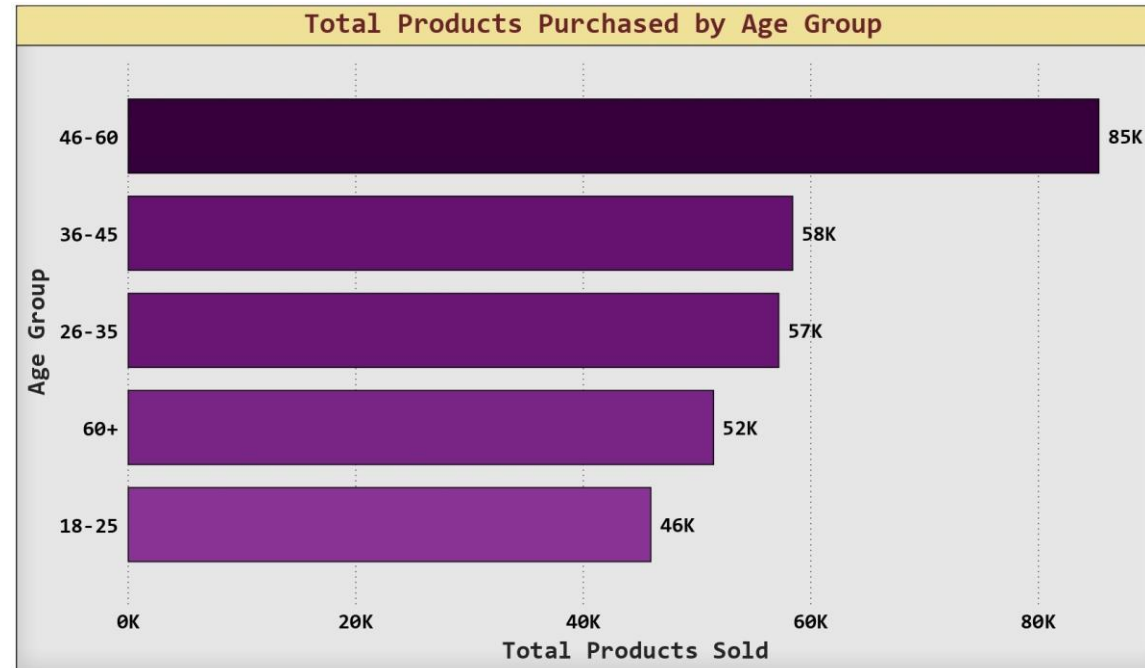
- Female revenue contribution is around 41M
- Male revenue contribution is around 28M
- Female segment generates higher overall revenue

Shopping Distribution by Age Group



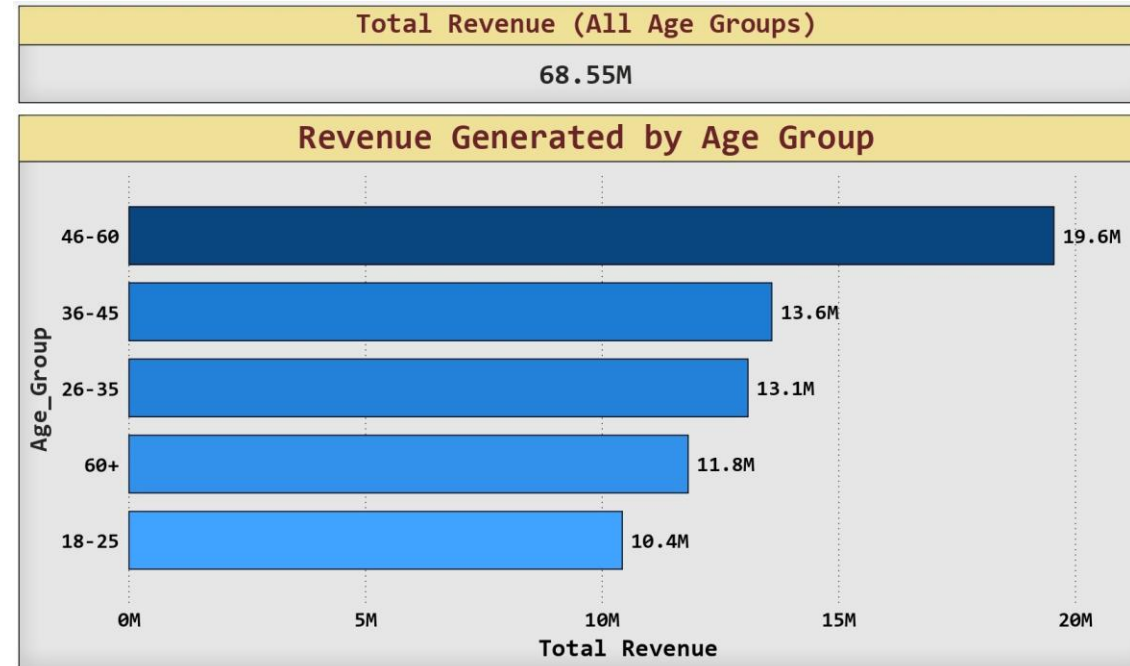
- Age group 46–60 shows highest transaction count
- 36–45 and 26–35 follow closely
- 18–25 and 60+ show relatively lower participation

Product Volume by Age Group



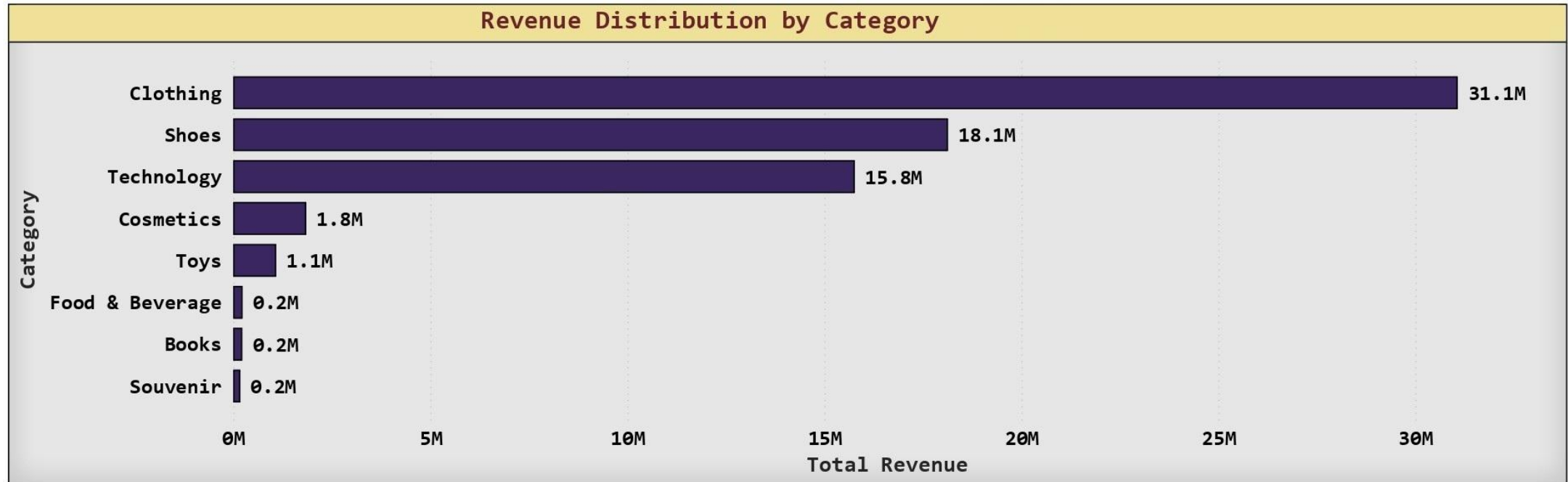
- 46–60 purchased the highest number of products (~85K)
- 36–45 and 26–35 show similar purchase levels
- Younger segment purchases fewer products comparatively

Revenue by Age Group



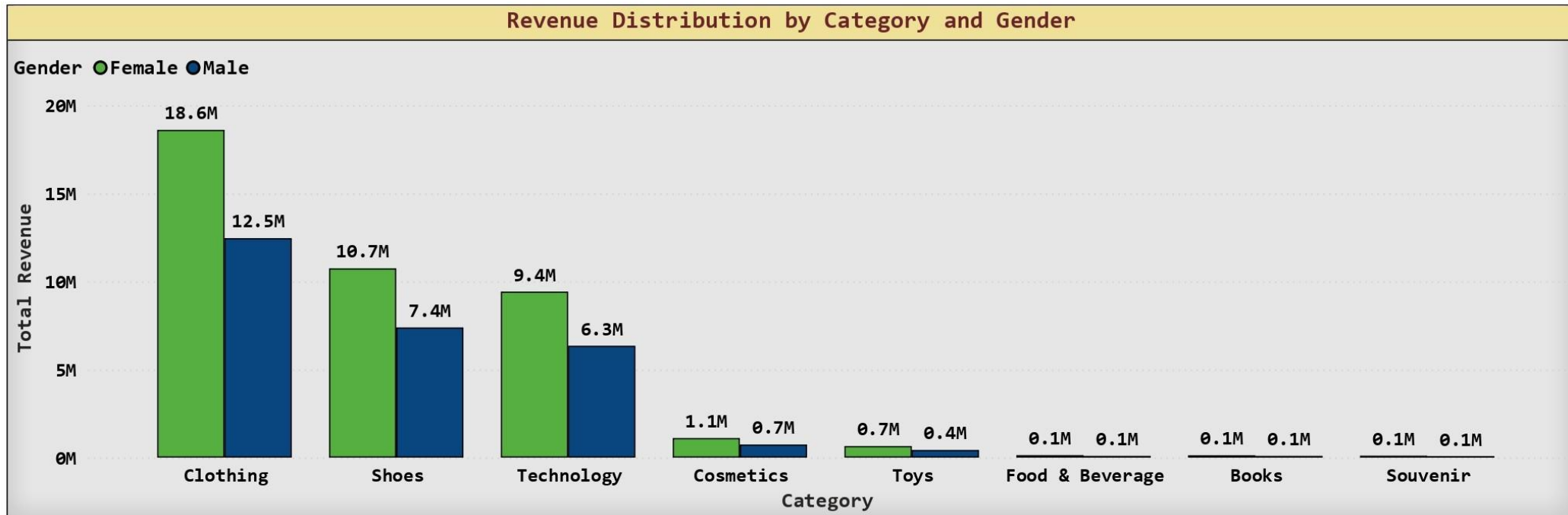
- 46–60 generates highest revenue (~19.6M)
- 36–45 and 26–35 are strong contributing groups
- Revenue decreases slightly for younger and senior segments

Category Performance Overview



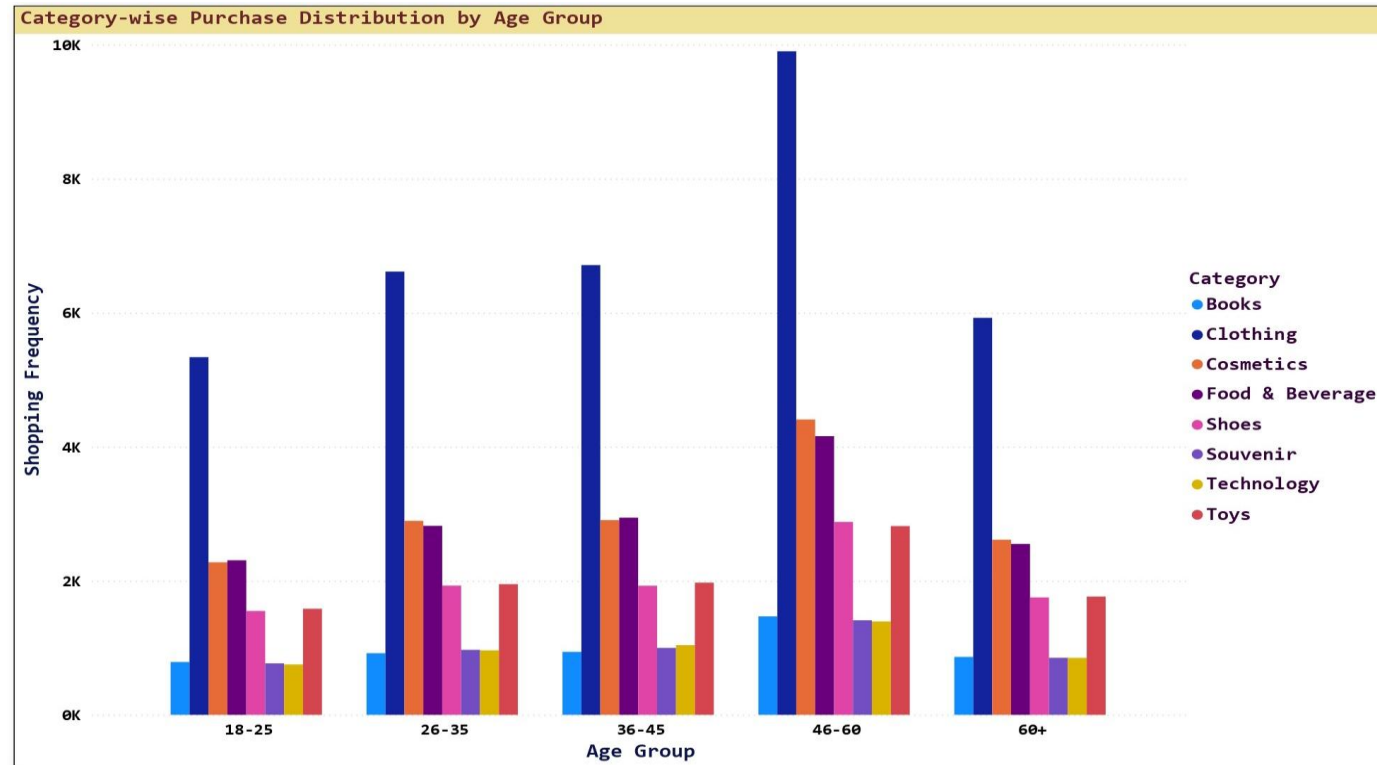
- Clothing is the highest revenue-generating category
- Shoes and Technology follow as strong contributors
- Books, Souvenir, and Food & Beverage contribute comparatively less

Category Insights by Gender



- Clothing revenue is higher for females than males
- Shoes and Technology also show strong female contribution
- Category demand pattern varies slightly across genders

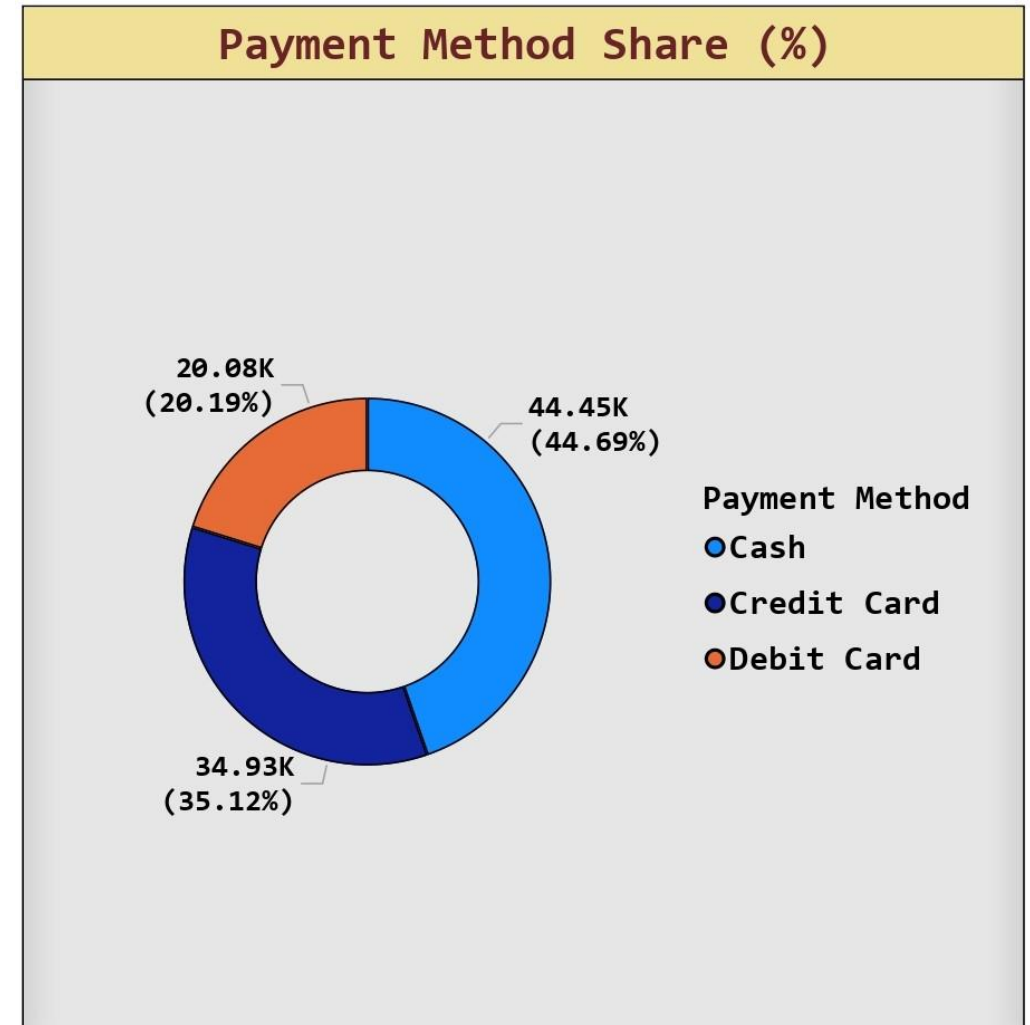
Category Insights by Age Group



- 46-60 shows strong revenue across major categories
- Clothing performs well across all age groups
- Technology and Shoes show consistent demand

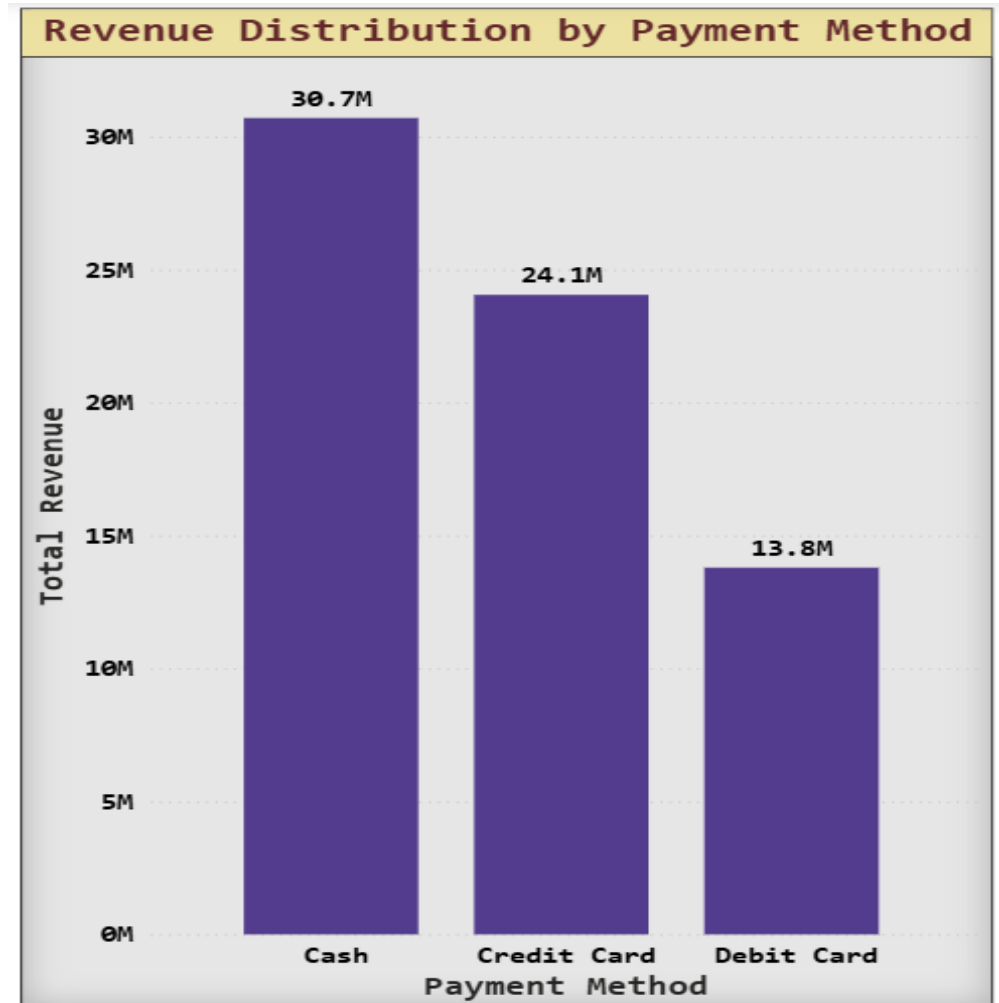
Payment Method Distribution

- Cash accounts for around 44.69% of transactions
- Credit card contributes around 35%
- Debit card contributes around 20%

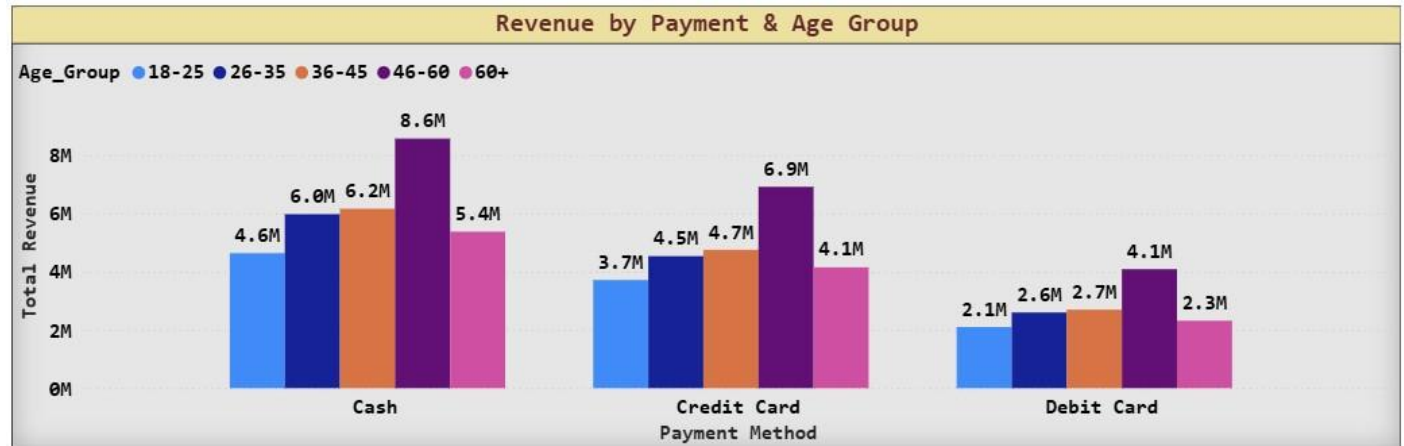
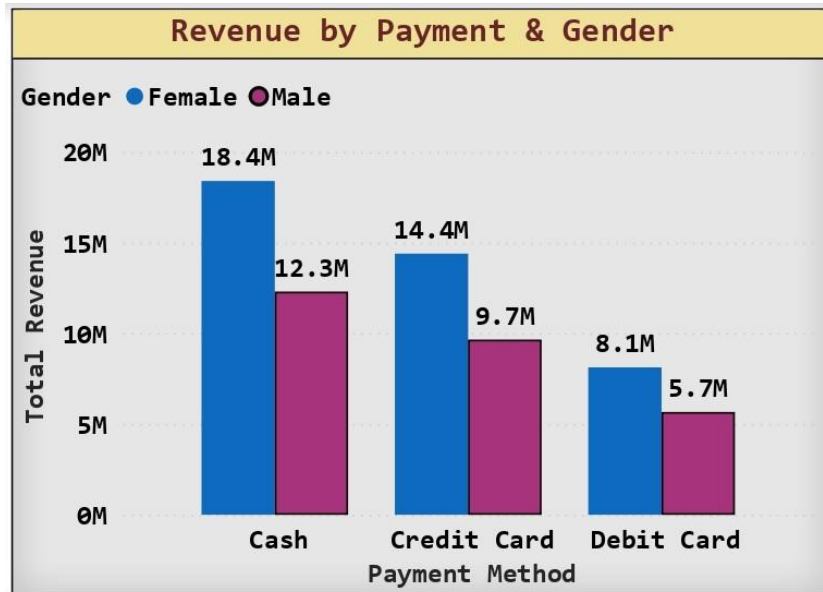


Revenue by Payment Method

- Cash generates the highest revenue
- Credit card follows closely
- Debit card shows lower revenue contribution



Payment Behavior by Gender and Age



- Female customers contribute more revenue across all payment types
- 46–60 age group shows higher revenue in all payment modes
- Digital payments are widely used but cash still dominates

Combined Key Insights

- Female customers are the strongest revenue-driving segment
- Age group 46–60 is the most valuable demographic
- Clothing, Shoes, and Technology drive major revenue
- Cash remains the most used payment method

Practical Business Recommendations

- Focus marketing campaigns on female and 46–60 segments
- Expand inventory in Clothing, Shoes, and Technology
- Encourage digital payments with targeted offers
- Introduce loyalty programs for high-value age groups

Limitations

- Dataset limited to selected malls and time period
- No customer-level repeat purchase tracking
- No profit margin data available for deeper analysis

Future Improvements

- Include customer segmentation using advanced analytics
- Add profit and cost data for margin analysis
- Perform predictive analysis for demand forecasting

Final Conclusion

- The analysis provides a clear view of customer behavior and revenue drivers
- Strong opportunities exist in targeted marketing and category expansion
- Data-driven decisions can improve long-term business performance